CourtIQ

An Intelligent Basketball Training System

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Mechanical Design

Rotating Plate

The positional servo motor is mounted on a wooden base platform and spins a rotating platform that tracks the player. The 10:1 gearbox supplies the necessary torque, while a 3D-printed flange securely connects the motor to the wooden structure. The system rotates freely on wheels, tracking the player across a 0–180° range.

Launch System

Two DC motors are mounted on the rotating platform and spin 3D-printed wheels with a rubber outer layer to grip the ball. High-density resin provides inertia to keep the wheels rotating smoothly during launch. An inclined V-shape wooden ramp funnels balls into the wheels for launch.

Electrical Design

The system is powered by a 48V, 1500W power supply, with two 0.15F supercapacitors to protect against current spikes caused by sudden motor load increase during ball launch. For motor control, the Jetson sends I2C signals to a PWM breakout board, which generates the required PWM outputs to enable the motors and control position/velocity. These signals control the servo position between 0 and 180°, while also regulating the DC motor speeds from 0 to 1400 rpm.

Problem

Many basketball players lack access to consistent training partners or coaches, limiting the efficiency and repetition of independent practice. Without a training partner, they are unable to recreate game-realistic practice drills to most effectively train.

Existing Solutions

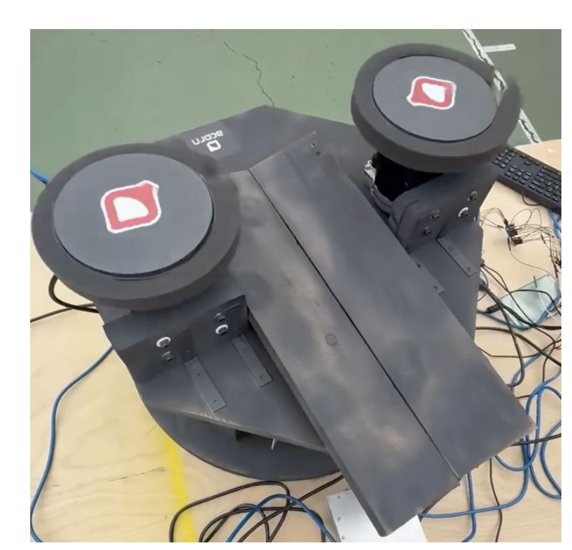
On the market, there are currently training systems available, such as the “Dr. Dish” However, these devices are expensive, bulky, and lack adaptability. These existing machines also solely shoot shots based on pre-programmed spots and lack real-time tracking, making it hard for players to customize their experience and shoot freely.

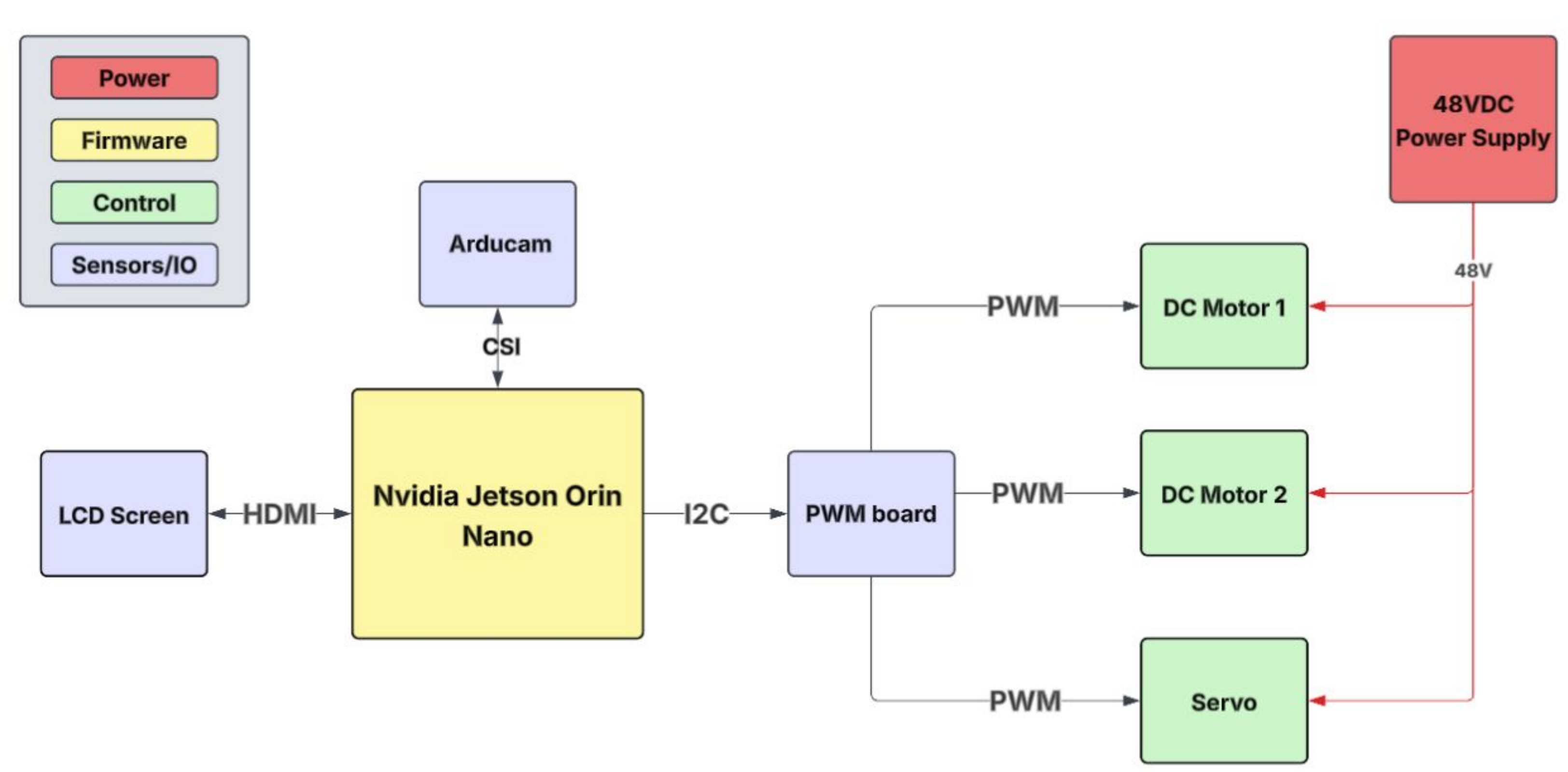
Our Solution

CourtIQ integrates computer vision for real-time player tracking, adapting to player movement and providing a more personalized and effective training experience. CourtIQ accomplishes:

1. Real-time position adjustment as player moves around the court.
2. Identifying when a player is ready for a shot.
3. Real-time adjustment of launch speed based on player distance from the hoop.
4. Handling multiple people in the frame and still tracking one player.

With this system, existing challenges are addressed by creating a dynamic, responsive training partner for ultimate solo skill development.



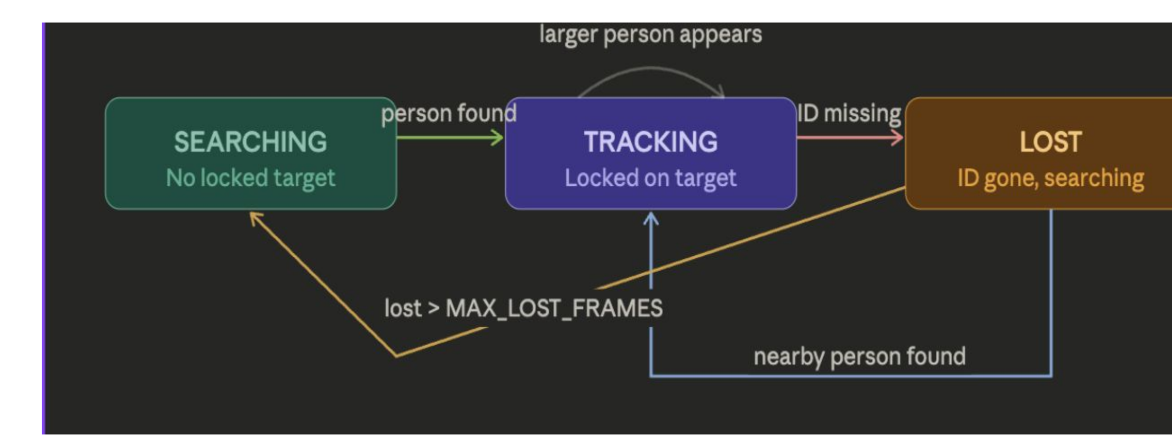


Software Design

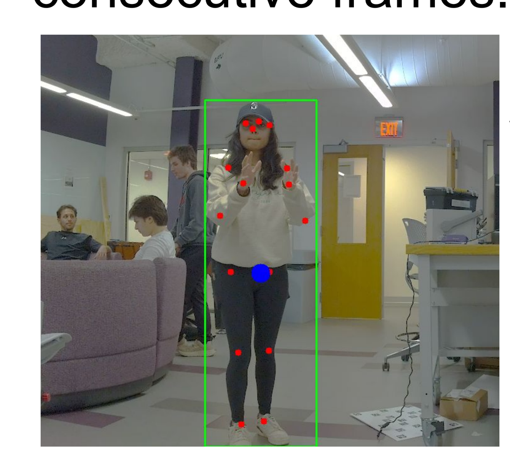
The software consists of three packages all written in **Python**. The packages are integrated together using QT, allowing communication of data between threads.

Computer Vision

Detects players in each frame using the **YOLOv8 Pose Model**, which outputs bounding boxes and pose keypoints. Tracking is implemented as a state machine with three states: *searching*, *tracking*, and *lost*.



During tracking, the player’s angle (0°–180°) is computed, and distance is estimated using the pinhole camera model. Shot detection tracks shoulder, elbow, and wrist keypoints, triggering when a “ready stance” is held across consecutive frames.



Player “ready stance” with bounding box and keypoints shown.

Pinhole Camera Model Distance Estimation Formula

FORMULA

$$d = \frac{f \cdot H}{h}$$

- d — distance to player
- f — focal length (px)
- H — player's real-world height (m)
- h — player's height in image (px)

Firmware

Receives angle and player distance from the CV, and maps these to PWM outputs to drive motor control.

duty = int((angle / 180.0) * 99)

pwm = int(duty / 100 * 0xFFFF)

User Interface

Allows players to input start/stop a session. Afterward, users can view statistics including total shots taken, shot locations on the court, and session duration.